AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3a; Solutions

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[1] Let $f\colon A \subseteq \mathbb{R}^n \to \mathbb{R}^m$.

[a] Let $B \subseteq \mathbb{R}^m$. Show that $f^{-1}(B^c) = A \cap (f^{-1}(B))^c$.

[b] Prove that f is continuous on A if and only if:

$$\text{for every } \textit{closed} \text{ subset } F \subseteq \mathbb{R}^m, \text{ then } f^{-1}(F) = A \cap E \text{ for some } \textit{closed} \text{ subset } E \subseteq \mathbb{R}^n. \quad (1)$$

(Note that if the domain A is all of $\mathbb{R}^n$, we can simplify this to $f^{-1}(F)$ is closed in $\mathbb{R}^n$ for every F closed in $\mathbb{R}^m$.)

[c] Show that $B = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$ is a closed subset of $\mathbb{R}^2$. *Hint:* Consider $g(x, y) = xy$.

Solution.

[a] We have

$$\begin{aligned} f^{-1}(B^c) &= \{\mathbf{x} \in A \mid f(\mathbf{x}) \in B^c\} = \{\mathbf{x} \in A : f(\mathbf{x}) \notin B\} \\ &= A \setminus f^{-1}(B) = A \cap (f^{-1}(B))^c. \end{aligned}$$

[b] We have already proved that f is continuous on A if and only if:

$$\text{for every } \textit{open} \text{ subset } V \subseteq \mathbb{R}^m, \text{ we have } f^{-1}(V) = A \cap U \text{ for some } \textit{open} \text{ subset } U \subseteq \mathbb{R}^n. \quad (2)$$

Thus we need only establish the equivalence of (1) and (2).

Suppose (1) holds. Let $F \subseteq \mathbb{R}^m$ be closed, so $V = F^c$ is open. Then $f^{-1}(V) = A \cap U$ for some open set U. By [a] we have $f^{-1}(F) = f^{-1}(V^c) = A \cap (f^{-1}(V))^c$. Thus, since $(f^{-1}(V))^c = (A \cap U)^c = A^c \cup U^c$, we get $f^{-1}(F) = A \cap (A^c \cup U^c) = A \cap (U^c)$. Since $E = U^c$ is closed, we deduce that (2) holds.

Suppose (2) holds. Let $V \subseteq \mathbb{R}^m$ be open, so $F = V^c$ is closed. Then $f^{-1}(F) = A \cap E$ for some closed set E. By [a] we have $f^{-1}(V) = f^{-1}(F^c) = A \cap (f^{-1}(F))^c$. Thus, since $(f^{-1}(F))^c = (A \cap E)^c = A^c \cup E^c$, we get $f^{-1}(V) = A \cap (A^c \cup E^c) = A \cap (E^c)$. Since $U = E^c$ is open, we deduce that (1) holds.

[c] The function $g\colon \mathbb{R}^2 \to \mathbb{R}$ given by $g(x, y) = xy$ is continuous on $\mathbb{R}^2$. Thus, since $F = \{1\} \subset \mathbb{R}$ is *closed*, by part [b] we get that $g^{-1}(F) = \{(x, y) \in \mathbb{R}^2 : g(x, y) = xy = 1\}$ is closed.

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[2] Let $f\colon A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a map. For $\mathbf{x} \in A$, we have

$$f(\mathbf{x}) = \big(f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})\big),$$

where $f_j\colon A \subseteq \mathbb{R}^n \to \mathbb{R}$ is the jth *component function* of f, for $1 \leq j \leq m$.

[a] Let $\mathbf{x}_0 \in A$. Prove using the sequential characterization of continuity that f is continuous at $\mathbf{x}_0$ if and only if each f_j is continuous at $\mathbf{x_0}$ for all $1 \leq j \leq m$.

[b] Let $\pi_j\colon \mathbb{R}^m \to \mathbb{R}$ be the map $\pi_j(y_1, \ldots, y_m) = y_j$, the projection onto the jth component. We proved in class directly using the definition of continuity that π_j is continuous on $\mathbb{R}^m$. Let $\iota_j\colon \mathbb{R} \to \mathbb{R}^m$ be the map $\iota_j(t) = (0, \ldots 0, t, 0, \ldots, 0)$ where the t is in the jth position. Prove directly using the definition of continuity that ι_j is continuous on $\mathbb{R}$.

[c] Find a way to use the maps π_j, ι_j in [b] to give another proof that f is continuous on A if and only if each f_j is continuous on A for $1 \leq j \leq m$.

Solution.

[a] Let $(\mathbf{x}_k)$ be a sequence in A converging to $\mathbf{x}_0$. Then we know that the sequence $(f(\mathbf{x}_k))$ in $\mathbb{R}^m$ converges to $f(\mathbf{x}_0)$ if and only if each of the component sequences $(f_j(\mathbf{x}_k))$ in $\mathbb{R}$ converges to $f_j(\mathbf{x}_0)$. Thus, by the sequential characterization of continuity, we conclude that f is continuous at $\mathbf{x}_0$ if and only if each f_j is continuous at $\mathbf{x}_0$.

[b] Let $t_0 \in \mathbb{R}$. We show that ι_j is continuous at t_0. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. If $|t - t_0| < \delta = \varepsilon$, then we have

$$\|\iota_j(t) - \iota_j(t_0)\| = \|(0, \ldots, 0, t - t_0, 0, \ldots, 0)\| = |t - t_0| < \varepsilon.$$

Thus ι_j is continuous at t_0.

[c] Observe that $f_j = \pi_j \circ f$, since $\pi_j(f(\mathbf{x}) = \pi_j(f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})) = f_j(\mathbf{x})$. Suppose f is continuous on A. Since π_j is continuous on $\mathbb{R}^m$, we have $f_j = \pi_j \circ f$ is continuous on A because the composition of continuous maps is continuous.

Observe that $f = \sum_{j=1}^m \iota_j \circ f_j$, since $\sum_{j=1}^m \iota_j(f_j(\mathbf{x})) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})) = f(\mathbf{x})$. Suppose each f_j is continuous on A. Since each ι_j is continuous on $\mathbb{R}$, we have $f = \sum_{j=1}^m \iota_j \circ f_j$ is continuous on A because sums and compositions of continuous maps are continuous.

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[3] Let $\mathbb{R}^{n^2}$ denote the space of $n \times n$ (square) matrices. Recall we can multiply two $n \times n$ matrices A, B to obtain a matrix AB.

[a] Let $A, B \in \mathbb{R}^{n^2}$. Recall that the Euclidean inner product on this space is $\langle A, B\rangle = \sum_{i=1}^n \sum_{j=1}^n A_{ij}B_{ij}$. Prove that $\langle A, B\rangle = \operatorname{tr}(A^T B)$, where A^T is the transpose of A, and $\operatorname{tr}(C)$ is the trace of C.

[b] Consider the map $g\colon \mathbb{R}^{n^2} \to \mathbb{R}^{n^2}$ given by $g(A) = A^T A$. Prove that g is continuous on $\mathbb{R}^{n^2}$.

[c] Prove that the subset of $n \times n$ *orthogonal matrices* is a *compact* subset of $\mathbb{R}^{n^2}$.

Solution.

[a] We have

$$\langle A, B\rangle = \sum_{i=1}^n \sum_{j=1}^n A_{ij}B_{ij} = \sum_{j=1}^n \Big(\sum_{i=1}^n (A^T)_{ji}B_{ij} \Big) = \sum_{j=1}^n (A^T B)_{jj} = \operatorname{tr}(A^T B).$$

[b] By Problem 2, the map g is continuous if and only if each component function $g_{ij}\colon \mathbb{R}^{n^2} \to \mathbb{R}$ is continuous. But

$$g_{ij}(A) = (A^T A)_{ij} = \sum_{k=1}^n (A^T)_{ik}A_{kj} = \sum_{k=1}^n A_{ki}A_{kj}$$

is a polynomial in the entries of A, and hence continuous.

[c] An $n \times n$ matrix A is *orthogonal* if and only if $A^T A = I$, where I is the identity matrix. Thus, the subset of orthogonal matrices is $g^{-1}(\{I\})$. Since a subset consisting of a single point is closed, $\{I\}$ is closed, and since g is continuous we conclude from part [b] of Problem 1 that $g^{-1}(\{I\})$ is closed. Thus, to show it is compact we need only show that it is bounded. But if A is orthogonal, then $A^T A = I$, so $\|A\|^2 = \langle A, A\rangle = \operatorname{tr}(A^T A) = \operatorname{tr}(I) = n$. Thus every $n \times n$ orthogonal matrix has norm $\sqrt{n}$, so this subset is also bounded.

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[4] Let $A \subseteq \mathbb{R}^n$ be nonempty, and fix a point $\mathbf{x} \in \mathbb{R}^n$.

[a] If A is *compact*, show that there exists at least one point $\mathbf{y} \in A$ such that $\|\mathbf{y} - \mathbf{x}\| \geq \|\mathbf{a} - \mathbf{x}\|$ for all $\mathbf{a} \in A$. That is, $\mathbf{y}$ is a point in A that is *furthest* from $\mathbf{x}$.

Hint: Find a way to use the Extreme Value Theorem.

[b] Give an example to show that $\mathbf{y}$ is not unique in general. (There are many possible answers.)

[c] If A is *not compact*, there may not exist any such $\mathbf{y}$. If A is noncompact because it is unbounded, then this is trivial to see, as there are points in A arbitrarily far from the origin, and hence arbitrarily far from any $\mathbf{x}$. More interesting is if A is noncompact because it is bounded but *not closed*. Give an example of such an A and $\mathbf{x}$ such that there is no farthest point in A from $\mathbf{x}$. (There are many possible answers.)

Solution.

[a] Consider the function $f\colon A \to \mathbb{R}$ given by $f((a)) = \|\mathbf{a} - \mathbf{x}\|$. This is clearly continuous, because it is the composition of the two continuous maps $\mathbf{a} \mapsto \mathbf{a} - \mathbf{x}$ and $\mathbf{z} \mapsto \|\mathbf{z}\|$. Thus, since A is compact, the Extreme Value Theorem says that there exists some $\mathbf{y} \in A$ such that $f(\mathbf{y}) \geq f(\mathbf{a})$ for all $\mathbf{a} \in A$.

[b] This "furthest point" in A to $\mathbf{x}$ need not be unique. For example, let $A = \{(s,t) \in \mathbb{R}^2 : s^2+t^2 = 1\}$. This is the "unit circle" in $\mathbb{R}^2$. It is compact because it is clearly closed and bounded. Let $\mathbf{x} = (0,0)$. Then any point on A is the same distance from $(0,0)$, so there are infinitely many points in A which are furthest from $(0,0)$.

[c] If A is bounded but not closed, then a "furthest point" need not exist. For example, let $A = B_1(\mathbf{0})$ be the open unit ball centred at the origin. This is bounded but not closed. Let $\mathbf{x} = \mathbf{0}$ be the origin. There is no point $\mathbf{a} \in A$ that is furthest from $\mathbf{0}$. If $\mathbf{a} \in A$ is nonzero, then $\|\mathbf{a}\| < 1$. Choose $t > 0$ such that $\|\mathbf{a}\| < \|t\mathbf{a}\| = t\,\|\mathbf{a}\| < 1$. (This is equivalent to $1 < t < \frac{1}{\|\mathbf{a}\|}$.) Then $t\mathbf{a} \in A$ and is further from $\mathbf{0}$ than $\mathbf{a}$.

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[5] Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be continuous on $\mathbb{R}^n$. Suppose that f has the following property: for any $\varepsilon > 0$, there exists $M > 0$ such that if $\|\mathbf{x}\| > M$, then $|f(\mathbf{x})| < \varepsilon$. (We say that f *goes to zero at infinity.*)

[a] Show that f is bounded on $\mathbb{R}^n$.

[b] Suppose there exists some point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f(\mathbf{x}_0) > 0$. Prove that there exists a point $\tilde{\mathbf{x}}$ such that $f(\mathbf{x}) \leq f(\tilde{\mathbf{x}})$ for all $\mathbf{x} \in \mathbb{R}^n$. (That is, show that f *attains a global maximum* on $\mathbb{R}^n$.)
[Note that we cannot *directly* use the Extreme Value Theorem because $\mathbb{R}^n$ is not compact. Find a creative way to use it nevertheless.]

[c] Show that the assumption that there exists $\mathbf{x}_0 \in \mathbb{R}^n$ with $f(\mathbf{x}_0) > 0$ is necessary for part [b]. That is, find an example where f *goes to zero at infinity*, with $f(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, and such that f does not attain a global maximum on $\mathbb{R}^n$. [Take $n = 1$ for simplicity.]

Solution.

[a] Take $\varepsilon = 1$. Then there exists $M > 0$ such that $\|\mathbf{x}\| > M$ implies that $|f(\mathbf{x})| < 1$. If $\|\mathbf{x}\| \leq M$, then $\mathbf{x} \in \overline{B_M(\mathbf{0})}$, which is compact. Since f is continuous, $f(\overline{B_M(\mathbf{0})})$ is compact, and thus bounded. So there exists $K > 0$ such that $\|\mathbf{x}\| \leq M$ implies that $|f(\mathbf{x})| \leq K$. Let $C = \max\{1, K\}$. Then $|f(\mathbf{x})| \leq C$ for all $\mathbf{x} \in \mathbb{R}^n$, so f is bounded on $\mathbb{R}^n$.

[b] Note that $f(\mathbf{x}_0) > 0$. There exists $M > 0$ such that $\|\mathbf{x}\| > M$ implies that $|f(\mathbf{x})| < \frac{1}{2}f(\mathbf{x}_0)$. In particular, this means that $\|\mathbf{x}_0\|$ must be less than or equal to M.

By the Extreme Value Theorem, on the compact set $\overline{B_M(\mathbf{0})}$, the function f attains a global maximum value $f(\tilde{\mathbf{x}})$ at some point $\tilde{\mathbf{x}} \in \overline{B_M(\mathbf{0})}$, and this global maximum value is at least $f(\mathbf{x_0})$, since $\mathbf{x_0} \in \overline{B_M(\mathbf{0})}$. On the complement of $\overline{B_M(\mathbf{0})}$, the function f takes on values that are strictly less than $f(\mathbf{x_0})$. Thus $f(\mathbf{x}) \leq f(\tilde{\mathbf{x}})$ for all $\mathbf{x} \in \mathbb{R}^n$.

[c] Take $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = -\frac{1}{(1+x^2)}$. Clearly f is continuous on $\mathbb{R}$, and $f(x) \leq 0$ for all $x \in \mathbb{R}$, but f does not attain a global maximum value on $\mathbb{R}$ because $\sup f(\mathbb{R}) = 0$ and $f(x) < 0$ for all $x \in \mathbb{R}$.
